

# SAURABH TRIPATHI

Senior Solutions Architect & Full Stack AI Engineer

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## PROFESSIONAL SUMMARY

Self-starter, results-oriented Full Stack AI Engineer with years of experience in building AI-powered applications, RAG pipelines, and multi-agent systems. Specialized in LangChain, LangGraph, vector search (FAISS), and LLM integrations (Gemini, OpenAI, Azure OpenAI). Expert in AI integrations with CMS and OTT platforms for enhanced editorial experience and user engagement. Proven track record of architecting AI-powered features and leading high-traffic media platforms serving millions of users. Triple certified CMS expert with expertise in full-stack development, cloud deployments (GCP, AWS), and technical team leadership. Adroit in communicating technical AI/ML concepts to business users and working in product environments with enterprise-level vendors.

## CERTIFICATIONS & COURSES

### Courses

- 📖 Antigravity and Gemini Enterprise — Google Cloud
- 📖 Architecting on AWS — Amazon Web Services

### Certifications

- 🏆 Triple Certified CMS Expert (Drupal 10, 11)
- 🏆 Triple Certified CMS Expert (Drupal 9)
- 🏆 Triple Certified CMS Expert (Drupal 8)
- 🏆 Certified CMS Developer (Drupal 7)

## TECHNICAL SKILLS

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|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------|
| <b>AI/ML:</b>                | LangChain, LangGraph, Google Gemini AI, OpenAI GPT, Azure OpenAI, Ollama, FAISS, RAG Pipelines, SentenceTransformers, Prompt Engineering | <b>Backend:</b>             | Python (FastAPI), PHP, Node.js, WebSocket, RESTful APIs, GraphQL, MySQL, PostgreSQL, Cloud SQL    |
| <b>Frontend:</b>             | React, TypeScript, JavaScript, Next.js, HTML5, CSS3, Twig, SCSS, jQuery, Tailwind CSS, marked.js                                         | <b>Vector &amp; Search:</b> | FAISS, SentenceTransformers, Vertex AI Search, Elasticsearch, Apache Solr, Semantic Search        |
| <b>Cloud (GCP):</b>          | Vertex AI, Cloud SQL (PostgreSQL), Cloud Run, BigQuery, Apigee, Log Explorer, Secret Manager, Cloud Build                                | <b>Cloud (AWS):</b>         | EC2, RDS, S3, CloudFront, Lambda, Cognito, Acquia Cloud                                           |
| <b>Data &amp; Databases:</b> | PostgreSQL, MySQL, MariaDB, Redis, BigQuery, Pandas, NumPy, ETL pipelines                                                                | <b>DevOps &amp; Tools:</b>  | Docker, Git, GitHub Actions, CI/CD, mitmproxy, Uvicorn, Poetry, uv, Pydantic, deepeval, pytest    |
| <b>Web Scraping:</b>         | Selenium 4, BeautifulSoup4, Playwright, mitmproxy, ChromeDriver (headless)                                                               | <b>Integrations:</b>        | SAML, OAuth, Payment Gateways, ADFS, BrightCove, Braze, AWS Cognito, Salesforce, Beyond Words TTS |
| <b>IDE &amp; Editors:</b>    | PhpStorm, VS Code, Sublime                                                                                                               | <b>Infrastructure:</b>      | Apache, LAMP, WAMP, Lando, DDEV Container, Acquia Cloud, Acquia Cloud Next, Pantheon              |
| <b>Development:</b>          | REST Web Services, Drupal/Drush console commands, Configuration sync, Unit testing, Agile/JIRA                                           | <b>Leadership:</b>          | Team Management, Problem Solving, Coaching & Mentoring, Teamwork, Collaboration                   |

## PROFESSIONAL EXPERIENCE

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### Senior Solutions Architect

June 2021 – Present

Mediacorp Pte Ltd

Singapore

- Architected and implemented AI-powered features: conversational search using RAG, multi-agent platforms with LangGraph, and LLM integrations
- Built Ask CNA AI search with FastAPI, Gemini 2.0, FAISS vector search, and WebSocket streaming for real-time responses
- Developed mewatch Conversational Platform using LangGraph, Vertex AI, Cloud SQL, and multi-agent orchestration on GCP Cloud Run
- Integrated AI with CMS and OTT platforms: title/description enhancement, content suggestions, and editorial workflow automation
- Implemented AI services including OpenAI modules, Beyond Words text-to-speech, and Vertex AI Search with custom RAG engines
- Lead team of 8 (backend, frontend, QA, support) and mentor on AI/ML best practices and modern architecture patterns
- Successfully executed CMS upgrades (v8→9→10→11) for multiple high-traffic platforms with AI integrations
- Developed email templating using Braze for enhanced user engagement and created AI-assisted development workflows
- Work with product owners to define features, review code, track projects, and drive delivery using DevOps tool stack
- Create technical documentation including API specifications, architecture diagrams, and RAG pipeline designs

### Software Engineer (Research and Development Team)

April 2019 – June 2021

Acquia India Private Limited

India

- Developed new plugins/modules for Drupal CMS (Acquia Product), automation scripts for testing and PSR4 standard reviews
- Led Drupal 7 to 8 migrations using Rector, Drupal Module Upgrader and active coding for complex module conversions
- Implemented API Integration and provided support/solutions to customers in setting up plugins
- Worked on multiple Drupal 8 websites under NDA with focus on site building, backend plugins, templates, hooks, services and debugging
- Handled migrations, Search API integrations, and worked on Acquia products: Site Studio, Cloud, Lift, CMS, and Machine Learning Dev tasks

### Application Development Senior Analyst

February 2017 – April 2019

Accenture

India

- Created chatbot using shell scripts and Python to automate developer workflows, configuration management, and deployment tasks
- Developed enterprise CMS applications for Fortune 500 clients with complex business logic and automation
- Integrated third-party services including authentication systems (SAML, OAuth), payment gateways, and RESTful APIs
- Resolved multilingual website support using i18n frameworks and created CLI commands for content migration and deployment
- Worked on government portal projects including EDF Energy portal (<https://www.edfenergy.com/>) with enterprise security and compliance requirements

## Software Engineer

October 2014 – February 2017

### Faichi Solutions

India

- Created Drubot chatbot using Artificial Intelligence Markup Language (AIML) for automated developer assistance and workflow optimization
- Developed enterprise CMS applications for Fortune 500 clients including Ixia (<http://www.ixiacom.com>, <https://support.ixiacom.com>) and Haivision (<http://www.haivision.com>)
- Implemented multilingual website support and created CLI automation commands for deployment and content management
- Integrated RSS/XML feeds, developed custom modules using open-source frameworks, and customized frontend themes
- Served as Full Stack Developer handling backend APIs, database optimization, and responsive frontend development

## Software Developer

January 2013 – September 2014

### Iksula

India

- Integrated e-commerce solutions into CMS platforms including payment gateway integration, shopping cart, and order management
- Developed custom modules using open-source frameworks for product catalog, inventory management, and promotional campaigns
- Worked with complex UI component libraries, data visualization, and batch processing APIs for bulk operations
- Implemented complete discount structure logic and promotional rules engine for books category for eBay.in project
- Integrated third-party APIs for shipping calculations, tax computation, and analytics tracking
- Developed projects including <http://www.bigbazaardirect.com>, <http://www.gitanjaligifts.com> and internal tools

## Software Engineer

June 2011 – December 2012

### Corpus Software

India

- Created shell scripts and PHP scripts for automated data import, content migration, and batch processing of CSV/XML files
- Developed responsive websites using PHP, MySQL, JavaScript, and CMS platforms with mobile-first design approach
- Built custom themes with responsive layouts and maintained enterprise client websites with version control
- Integrated third-party services including email systems, analytics platforms, and social media APIs
- Provided technical support and conducted training sessions for government corporations on ERP workflows and CMS usage

## KEY PROJECTS

### Ask CNA - AI Search — AI Engineer & Solutions Architect

AI-powered conversational search for CNA using RAG pipeline. Built with FastAPI, Gemini 2.0, FAISS vector search, and WebSocket streaming. Real-time answer generation with article citations, sub-2-second response time, semantic ranking using SentenceTransformers. Python, FastAPI, Gemini AI, FAISS, Selenium, mitmproxy, WebSockets, BeautifulSoup

### CNA, Berita & Seithi — Lead Solutions Architect

Major media platforms (CNA, Berita, Seithi) serving millions daily. Integrated AI for CMS and OTT: editor experience enhancement (title/description optimization), OpenAI modules, Beyond Words TTS. 99.9% uptime, multiple version upgrades (v7→11). CMS, React, TypeScript, AI integrations, AWS, ADFS, BrightCove

### Ixia & Haivision — Software Engineer

### mewatch Conversational Platform — AI Engineer

Multi-agent conversational platform for OTT content discovery. Built with LangGraph supervisor orchestration, Vertex AI Search, Cloud SQL memory, and RAG engine for football rules/FAQs. Deployed on GCP Cloud Run with semantic versioning. LangGraph, LangChain, Vertex AI, Cloud SQL, GCP Cloud Run, Apigee, FastAPI

### EDF Energy Portal — Senior Analyst

UK government energy portal (<https://www.edfenergy.com/>). Supported multiple features with contributed/custom modules and complex business logic. Extended contributed module to create chatbot for automating developer work. Drupal 8, Custom modules, Chatbot integration

Enterprise Drupal 8 solutions for <http://www.ixiacom.com>, <https://support.ixiacom.com>, and <http://www.haivision.com>. Handled multilingual support, Drush commands, feed integration and full stack development.

Drupal 7/8, Multilingual, Feed integration

## AWARDS & ACHIEVEMENTS

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- Digital Hackathon first prize for creating AI-powered conversational search (Ask CNA)
- Module Contribution: Maintainer of multiple projects on drupal.org
- Multiple Digi Commercial Awards and Voyager awards at Mediacorp
- ACE (Accenture Celebrates Excellence) award at Accenture
- Star of the month at eBay.in in Books category working onsite
- Successfully demonstrated Drupal application to Indian State Government
- Conducted 1 week training to Government Corporations on using ERP application workflows in Drupal

## OPEN SOURCE CONTRIBUTIONS

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**Drupal.org Profile:** [drupal.org/u/saurabhtrpathics](https://drupal.org/u/saurabhtrpathics) 13+ years on Drupal.org | 92 issue credits | 4 Drupal Core contributions

**Projects Maintained:** AIML Parser, Bulk delete 301, Country Specific Nodes, D.A.I.L., drubot, PasswordResetTabs

**Notable Contributions:** Drupal Core, Chaos Tool Suite (ctools), Index Now, Contact Storage, Feeds, Lightning Media, Schema.org Metatag

## EDUCATION

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Bachelor of Technology — Gautam Buddh Technical University, 2011